

RESEARCH ARTICLE

FARE: Financial Agentic Reasoning and Evaluation for Earnings Call Transcripts

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ABSTRACT Earnings call transcripts contain crucial information about the financial performance of companies across various industries. However, due to their unstructured nature, an automated interpretation system is challenging to build. This study presents Financial Agentic Reasoning and Evaluation (FARE) framework that analyzes earnings call transcripts and answers users' queries using an integration of retrieval-augmented generation (RAG) and multi-agent reasoning. The framework built using LangGraph contains six agents: Statement Extraction, Tone, Investment, Comparison, Tone Shift, and Simulation. Each agent performs a specialized task ranging from providing investment guidance to analyzing company sentiment. Due to the lack of evaluation metrics for assessing performance of agentic systems with earnings call transcripts, we introduced evaluation metrics to evaluate each agent using an LLM-as-a-judge system. The FARE framework was evaluated on a curated dataset of 168 quarterly earnings call transcripts from 21 companies. We compared the performance of four LLM backbones: Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Qwen2.5-7B-Instruct, and Llama-3.2-3B-Instruct which were integrated in the system. The results show that Qwen2.5-7B-Instruct achieves the strongest overall reasoning performance across multiple agents, while Gemini 2.5 Flash-Lite consistently demonstrates significantly lower latency and higher efficiency.

INDEX TERMS Agentic AI, financial reasoning, earnings call transcripts, retrieval-augmented generation, large language models, LangGraph, reasoning alignment.

I. INTRODUCTION

Earnings call transcripts provide a glimpse into the perspectives of the executives on quarterly results and future strategies. The transcripts comprise dialogues between chief executives, analysts and a moderator. The discussions are not only about revenue figures and earnings per share, but also about the future outlook of the company based on optimistic projections and competition from companies. The number of these discussions and disclosures are extremely high, often exceeding 10,000 annually from S&P 500 firms alone. Traditional qualitative analysis methods have found it difficult to rely on human interpreters to extract meaningful signals and information from the large volume of data, leading to delayed insights and incorrect

interpretations [1]. Due to the dominance of algorithmic trading and high-frequency decision-making, the demand for scalable and automated tools to process these texts has rapidly increased. Users want tools that are capable of understanding the subtleties of spoken discourse, which include rhetorical flourishes, conditional phrasing, and cross-referential cues that avoid a simple keyword matching algorithm.

The addition of artificial intelligence (AI) into the financial domain has caused a shift from rule-based classifiers in the early 2010s to the current generative paradigms that utilize large language models (LLMs). Early applications focused on sentiment detection, using dictionaries to classify texts for bullish or bearish tones. However, these approaches failed against the contextual intricacies of financial language, such as hidden signals and industry jargon [2]. Transformer-based architectures that were fine-tuned on financial corpora

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achieved reasonable gains in correlating textual signals with asset price reactions [3]. More recently, industries have pivoted towards agentic systems which utilize multi-step workflows and invoke external tools [4]. These agents utilize retrieval mechanisms to ground their outputs in factual data, allowing the systems to shift from traditional text summarization to active reasoning and analysis of financial texts.

Despite the rise of agentic AI systems, there is limited application of such systems on earnings calls. Unlike regular financial text, earnings call transcripts provide a unique set of challenges for frameworks. The structure of the dialogues blends prepared remarks with impromptu Q&A, and historical references are scattered across the calls. It is vital for frameworks to avoid errors while inferring, since that can amplify market distortions [5]. Most previous studies emphasized utilizing isolated components for different tasks, such as classifying sentiment on earnings call transcripts [6]. The lack of an end-to-end framework reduces the generalization ability of AI models on real-world finance data. Moreover, the systems do not include robust evaluation frameworks tailored for financial decision-making.

While prior studies and research has applied traditional natural language processing (NLP) techniques and simplified agentic systems to financial texts, limited work has explored a multi-agent reasoning framework that integrates retrieval-augmented generation (RAG), LLMs and domain-specific evaluation metrics for analyzing earnings call transcripts. To address these gaps, this paper makes the following key contributions:

- We propose FARE, a novel multi-agent framework for earnings call transcript analysis. The framework integrates RAG with agent-based reasoning to enable fact-grounded and interpretable analysis.
- We construct a comprehensive dataset of 168 quarterly earnings call transcripts across 21 companies. Each transcript is systematically processed through segmentation, semantic embedding, and Facebook AI Similarity Search (FAISS)-based vector indexing. This processing allows for scalable and efficient retrieval of contextually relevant information during agent reasoning.
- We design six specialized agents including Tone, Investment, Tone Shift, Simulation, Comparison, and Statement Extraction. Each agent performs domain-specific reasoning tasks that are routed using an Orchestrator Agent within a coordinated LangGraph workflow. The Orchestrator Agent interprets the user intent and dynamically routes the queries, ensuring deterministic execution of the workflow.
- We introduce a novel set of evaluation metrics including Tone Alignment Evaluation Framework (TAEF), Tone Alignment Score (TAS), Investment Reasoning Alignment Score (IRAS), Evaluation Alignment Score (EAS), and Comparison Alignment Score (CAS). These metrics were designed to judge contextual alignment,

interpretive reliability, and reasoning robustness across the agents.

- We compare the performance of FARE on the evaluation metrics and latency using four models, namely Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Qwen2.5-7B-Instruct, and Llama-3.2-3B-Instruct as LLM backbones. The evaluation analyzes agent-level reasoning quality, retrieval alignment, and system latency. The framework provides insights into the trade-offs between the reasoning performance and efficiency in a multi-agent financial system.

The results demonstrated the effectiveness of the proposed framework in interpreting earnings call transcripts using an agentic RAG pipeline, while maintaining low-latency performance. The remainder of this paper is organized as follows: Section II provides a comprehensive review of related work on NLP techniques, RAG architectures, and multi-agent systems in the financial domain. Section III describes the process of dataset collection, preprocessing, and vector database construction. Section IV outlines the overall construction of the FARE framework, providing detail for each of the six specialized agents, and the novel evaluation metrics used to assess their performance. Section V compares the performance between the four LLM models across all six agents. Section VI discusses the potential risks associated with the proposed framework and the corresponding mitigation strategies. Section VII addresses the limitations of FARE and future enhancements to improve it. Finally, Section VIII concludes the paper.

II. RELATED WORK

The rapid evolution of artificial intelligence and natural language processing has transformed the landscape of financial text analysis. From sentiment extraction and market prediction to agentic reasoning systems, research has advanced toward more autonomous and interpretable financial intelligence. Traditional econometric and lexicon-based techniques have given way to deep learning architectures capable of capturing linguistic nuance and context from large-scale textual data such as earnings call transcripts and analyst reports. More recently, retrieval-augmented generation and multi-agent frameworks have begun redefining how financial narratives are interpreted, shifting the focus from static prediction toward dynamic reasoning, explainability, and cross-source evidence integration.

A. NLP AND SENTIMENT ANALYSIS IN FINANCIAL TEXTS

Sentiment extraction from financial texts like transcripts has progressed from lexicon-based methods to advanced NLP. Grammar-aware engines parse sentences for collective sentiments and polarity, outperforming basics on corpora and linking moods to stock indices via Granger causality [2]. Deep learning models such as CNN-LSTM gave an AUC score of 0.74 when fine-tuned on multimodal earnings call transcripts [1]. Further advancements included hierarchical

attention mechanisms to weigh contextual dependencies in earnings calls. This enabled the detection of subtle shifts in the tone of executives that correlate with intraday price volatility. The changes in the architecture of the models were demonstrated in the analysis of S&P 500 firms where sentiment granularity improved forecast precision by 12% over aggregate scoring [7].

LLM benchmarks exposed gaps in financial nuance, like hedged statements, with models such as GPT-4o having accuracies under 85% [3]. Domain-adapted prompting is important for reducing errors when encountering financial terminology and industry jargon [8]. Recent explorations into zero-shot learning for earnings call transcripts improved the performance of instruction-tuned models in identifying forward-looking risks without labeled data. These models achieved F1-scores of 0.82 on implied uncertainty phrases. However, challenges still persist in cross-sector generalization due to varying linguistic conventions [9].

B. AI AGENTS AND RETRIEVAL-AUGMENTED GENERATION IN FINANCE

Agentic AI systems integrated with RAG address LLM gaps through the use of tools and self-refinement. The usage of reflection for financial tasks like market queries is becoming increasingly important, along with multi-step retrieval systems [10]. RAG systems using OpenAI embeddings and GPT-4 scored 3.7/5 when tested on bank reports; however, these systems performed poorly with numerical PDFs, suggesting a hybrid system would be better [11]. Innovations like the earnings conference call (ECC) Analyzer extend this to earnings calls by employing a two-stage RAG process: paragraph-level summarization followed by query-driven sentence retrieval from a predefined investor question bank, fusing text and audio features (via SimCSE and Wav2vec2) to predict stock volatility with a 27.7% MSE reduction over state-of-the-art baselines across 3- to 30-day horizons [12].

RAGAS enables reference-free evaluation of faithfulness (95% human alignment) on event datasets, auditing risk analysis where contexts boost reliability but reveal adversarial weaknesses [13]. Emerging hybrid agents integrate graph-based retrieval for relational financial data, such as linking entity mentions in reports to external market graphs, enhancing QA accuracy on complex queries involving causal inference, while mitigating over-reliance on parametric knowledge through dynamic reranking.

C. MULTI-AGENT SYSTEMS FOR FINANCIAL DECISION-MAKING AND TRADING

Multi-agent systems emulate trading teams with role assignments. TradingAgents uses LLM debates on sentiments and indicators for 23–26% returns and Sharpe 8.21, exceeding rules via explanations [14]. Its architecture incorporates specialized teams, including analysts focusing on fundamentals, sentiment through social media data, news, and

technical indicators; bullish and bearish researchers for debate; traders for execution decisions; and risk managers with varying tolerance levels. This multi-role design achieved cumulative returns (CR) of up to 26.62% on technology stocks such as AAPL (Apple Inc.) during backtesting, with maximum drawdowns remaining below 2%, supported by structured ReAct prompting and reasoning through the o1-preview model [14]. TradingGPT adds layered memories with decay for risk-based debates on multimodal data, aiding volatility [15].

FINCON models manager-analyst interactions with reinforcement, surpassing deep reinforcement learning (DRL) in optimization and multi-asset generalization via value-at-risk [16]. Ontology-based systems decouple knowledge for simulations, pausing trades in events like mergers [17], [18]. FinRobot orchestrates LLMs for chain-of-thought in forecasting, synthesizing earnings with real-time data for predictions like NVDA (NVIDIA Corporation) movements [19]. Multi-agent DRL on fractal hypotheses improves forex returns through the use of hierarchical flows [20]. Vertical ensembles optimize risk tasks on filings, achieving an accuracy of 66.7% [21]. Collaborative frameworks were used to further explore persona-driven negotiations. These frameworks had agents simulate institutional investors to resolve conflicting signals from earnings transcripts, boosting portfolio diversification scores by 18% in simulated bull-bear cycles [22].

D. EVALUATION AND RISK FRAMEWORKS FOR AGENTIC SYSTEMS

Agentic AI financial systems require auditing for safety. In this context, safety refers to ensuring that the system produces factually reliable and consistent outputs while minimizing hallucinations and unsupported claims. It also involves maintaining coherent multi-step reasoning and preventing misleading or high-risk financial interpretations that could negatively impact decision-making.

To address these challenges, prior works have proposed various evaluation and risk-aware frameworks. A Multi-agent, Safety-Aware Evaluation Agent (M-SAEA) is used to detect hallucinations and fragilities in trajectories, scoring 75% on crypto simulations [5]. Surveys on LLM trading note 15–30% backtests, but stated hybrid systems would be ideal for latency and ethics [4]. The FINCON framework introduces a dual-level risk control system, using conditional value-at-risk (CVaR) for within-episode alerts on daily profit and loss (PnL), and conceptual verbal reinforcement for over-episode belief updates. The framework achieved cumulative returns up to 113.8% in portfolios with Sharpe ratios crossing baselines by 2x and drawdowns minimized through timeliness-aware memory decay [16].

Human-AI loops also started being used to build trust with users [6]. The use of tools to extract expert insights on earnings calls was also explored, but scaling the systems remained a challenge [1]. Comprehensive risk suites started

incorporating trajectory probing for multi-agent finance systems, flagging temporal inconsistencies in decision chains with 82% precision [5], [13].

Overall, previous studies revealed a clear progression from sentiment analysis of financial texts to integrating RAG with agentic systems for proper decision-making workflows in financial contexts. However, most frameworks used isolated components such as tone, reasoning, or simulation without unifying them under a comprehensive agentic framework. FARE aims to build upon these advancements by combining RAG, multi-agent reasoning, and robust evaluation metrics to create a cohesive system for financial interpretation. The proposed framework aims to answer a diverse set of questions from users and provide fast agent responses that are grounded in factual data.

III. DATA COLLECTION AND DATASET CONSTRUCTION

This section dives into the process of procurement of the earnings call transcripts, the preprocessing pipeline, and the embedding techniques to be used for the FARE framework. All the earnings call transcripts were obtained from official investor relations websites. Twenty-one companies were selected that spanned diverse sectors such as technology, consulting, finance, and retail. For each company, only the eight most recent transcripts were used since we did not want outdated data. We wanted enough data to show market changes over time, as well as recent data so that it reflects the current market conditions.

A. RAW DATA ACQUISITION

Figure 1 illustrates the pipeline of dataset construction. All the quarterly transcripts from the companies were originally in PDF format. Each transcript was converted to regular text using the PyMuPDF library. After that, all the text was cleaned by removing non-semantic layout artifacts such as page boundaries, headers, footers, and irregular spacing. We ensured that all relevant content, such as speaker labels of CEO, CFO, and section boundaries such as Prepared Remarks and Q&A, were preserved. Minimal cleaning was applied to ensure that the natural structure and meaning of the transcripts were not distorted.

B. PREPROCESSING AND CHUNK STRUCTURING

As mentioned earlier, each transcript was normalized by removing unnecessary whitespaces and headers and footers. Then, in order to reduce computational load, each transcript was segmented into meaningful chunks. These chunks preserve the original content of the document while reducing the burden of large storage. This is crucial for RAG-based systems because LLMs have a limited context window and can cause hallucinations.

The chunking of the earnings call transcripts was done using a recursive character-based splitting strategy. The size of each chunk was 800 characters, with an overlap of 100 characters between chunks to ensure that the semantic meaning across chunk boundaries was preserved.

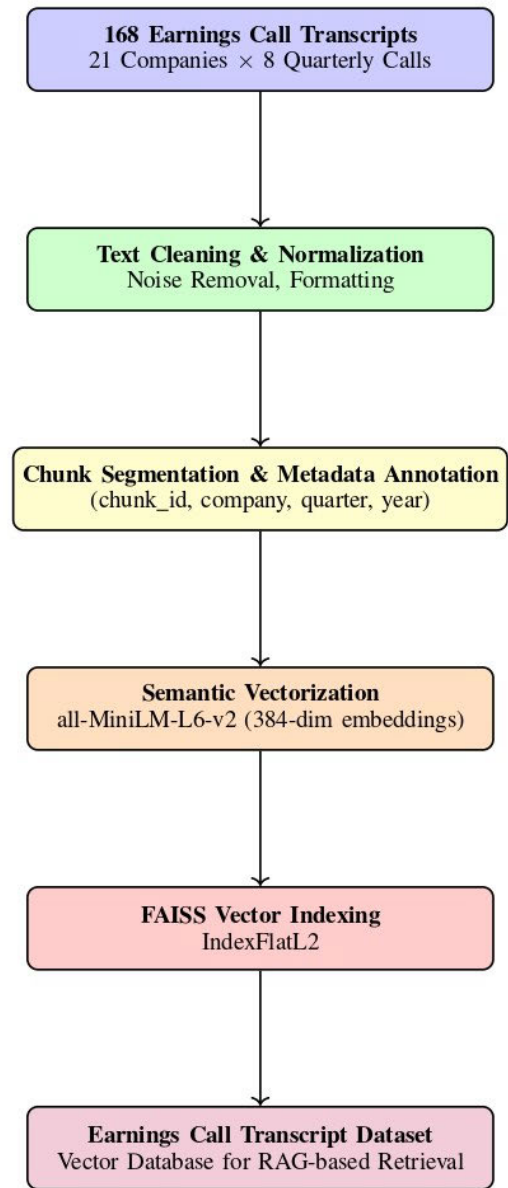


FIGURE 1. Pipeline of dataset construction depicting the stages of preprocessing, chunking, embedding, and FAISS storage.

A hierarchical separation strategy was employed, starting from paragraph breaks and line breaks up to whitespaces. Since earnings call transcripts involve conversations and discussions about the company, a lot of information is spread across chunks. Therefore, it is crucial to avoid loss of data, which is why a recursive splitting strategy was utilized.

Although advanced semantic or dialogue-aware chunking strategies could further preserve conversational coherence, such approaches require additional structural parsing and significantly increase computational overhead. Such approaches were outside the scope of this study, so a fixed chunk size of 800 characters with a 100-character overlap was used to balance semantic context preservation with computational efficiency.

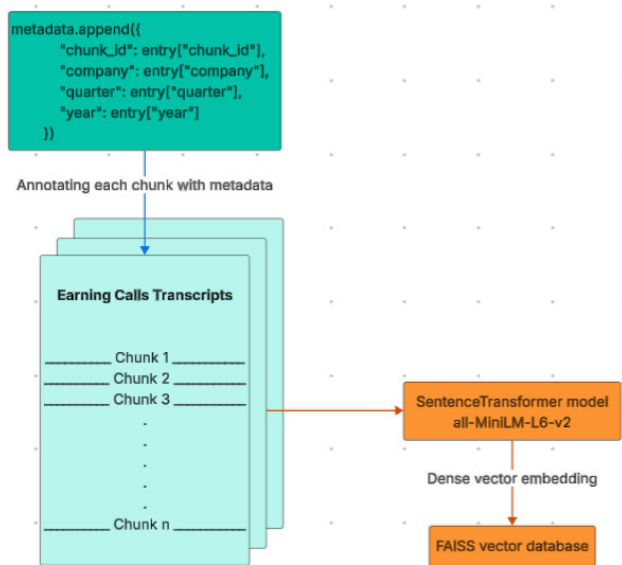


FIGURE 2. Overview of the metadata annotation of the transcript chunks and the embedding workflow.

To ensure that downstream LLMs can easily retrieve the correct transcript chunks, each chunk was annotated with metadata. Figure 2 provides an overview of the metadata annotation and embedding workflow. The annotated metadata included the chunk_id, company, quarter, and year.

C. SEMANTIC EMBEDDING AND VECTOR INDEXING

The final stage of the dataset construction is the storage of the chunks in the vector database. All the chunk texts were transformed into dense vector embeddings using the all-MiniLM-L6-v2 model from the SentenceTransformers library. This model maps paragraphs and sentences into a 384-dimensional embedding. For 168 transcripts, this embedding choice provides a balance between strong semantic representation and computational efficiency, making it suitable for scalable and interactive financial question answering.

Each of these embeddings was then indexed using FAISS IndexFlatL2, which enables exact nearest-neighbor searching using L2 distance. Rather than using approximate methods, we wanted an exact search algorithm for chunk retrieval that aligns well with the embedding space generated by the MiniLM model, which is why the L2-based strategy was utilized.

D. DATASET COMPOSITION

All the chunk embeddings were stored in FAISS (Facebook AI Similarity Search). FAISS provides efficient similarity search and clustering of high-dimensional vectors. Since we want fast retrieval of chunks for the FARE framework, FAISS was chosen.

Table 1 summarizes the companies and their respective industries used in the consolidated dataset. A total of 168 earnings call transcripts spanning diverse sectors

TABLE 1. List of companies and their corresponding industries.

No.	Company Name	Industry
1	3M [23]	Manufacturing
2	Accenture [24]	Consulting and Technology Services
3	Alphabet [25]	Technology
4	Boeing [26]	Aerospace and Defense
5	Citibank [27]	Financial Services
6	ExxonMobil [28]	Energy
7	Ford [29]	Automotive
8	John Deere [30]	Industrial and Agricultural Equipment
9	JPMorgan [31]	Financial Services
10	Microsoft [32]	Technology
11	Moderna [33]	Biotechnology
12	Netflix [34]	Media and Entertainment
13	Nike [35]	Apparel and Footwear
14	PepsiCo [36]	Consumer Goods
15	Pfizer [37]	Pharmaceuticals
16	Salesforce [38]	Software and Cloud Computing
17	Starbucks [39]	Food and Beverage
18	Target [40]	Retail
19	Uber [41]	Transportation and Technology
20	Visa [42]	Financial Technology
21	Walmart [43]	Retail

was selected, ensuring that the framework is suitable for multi-purpose users and stakeholders.

IV. METHODOLOGY

This section presents the comprehensive methodology of the proposed FARE framework. Figure 3 illustrates the workflow of the framework. A user will ask a question to the main orchestrator agent. The question can range from the tone of a transcript to which company to invest in. The Orchestrator Agent will comprehend the question and analyze which specialized agent the query corresponds to. From the Agents Layer, which consists of the six specialized agents used in this framework, an agent is routed by the Orchestrator Agent.

The Orchestrator Agent utilizes a single-intent routing strategy, where each user query is mapped to one predefined task category using an LLM-based routing prompt. So, for each user query, only one specialized agent is activated to ensure deterministic and interpretable system behavior. In situations where the user query is ambiguous, the Orchestrator Agent relies on the large language model’s semantic understanding to infer the dominant intent and route the query accordingly. This design choice reduces the complexity of the framework and avoids conflicting agent outputs.

The specialized agents are the Tone Agent, Investment Agent, Tone Shift Agent, Simulation Agent, Comparison Agent, and Statement Extraction Agent. The details of these agents will be further explored in the following subsections. In general, each of the agents must communicate with the FAISS vector store, which contains the embedded chunks of the 168 transcripts.

The framework integrates a retrieval-augmented generation (RAG) backbone, where the selected agent searches for the nearest and most relevant context to the question

from the transcripts, and then receives the retrieved and filtered chunks. Each of the six agents and the Orchestrator Agent is powered by an LLM Generation Backbone, meaning that Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Qwen2.5-7B-Instruct, or Llama-3.2-3B-Instruct is used as the underlying LLM. We compare the performance of both these LLMs across all agents using novel evaluation metrics and latency. Finally, the generated and synthesized answer from the agent is returned back to the user. The entire workflow is developed using LangGraph, an open-source multi-agent workflow system.

A. AGENTIC FRAMEWORK DESIGN

The workflow of FARE is designed using LangGraph. LangGraph provides the orchestration layer that governs the multi-agent architecture. Each node is represented by an agent, and each edge represents the connection between agents. LangGraph contains a state component, which was crucial for the design of the framework. Information such as the agent's response and the user portfolio was propagated from the orchestrator to the agents without any loss of data. Inter-agent coordination is governed through a centralized orchestrator with single-intent routing, ensuring deterministic task execution. Logical consistency across multi-step reasoning is maintained through structured prompts and controlled state propagation between agents. Each of the six specialized agents was evaluated by six separate LLM evaluator agents using the LLM-as-a-judge technique. The advantage of FARE compared to previous frameworks is that each of the six agents has domain-specific tasks and follows a modular design. They are capable of handling a large number of transcripts and can understand the system prompts designed for them.

The framework supports batch-wise analysis of the transcripts due to the limited context window of the LLM backbones. We ensured that the batch size, the top-k chunks, and other hyperparameters were consistent across all the agents and during the comparison evaluation.

In our implementation, the retrieval process uses a fixed top-k setting of $k = 10$, where the top 10 most semantically similar chunks are retrieved from the FAISS vector store for each query. This value was selected to balance retrieval coverage and contextual completeness for downstream reasoning. All non-Gemini models, including Qwen2.5-7B-Instruct and Llama-3.2-3B-Instruct, were executed on a single NVIDIA T4 GPU using Google Colab, while Gemini-based models were accessed via API-based cloud inference.

The prompting strategy across all agents follows a role-based design, where each agent is assigned a specific functional role (e.g., Tone Analyst, Investment Advisor). Additionally, Chain-of-Thought style prompting is employed to decompose complex financial queries into structured reasoning steps, enabling the agents to perform multi-step analysis such as temporal aggregation, evidence synthesis, and decision justification.

In this study, the framework is evaluated using four LLM model backbones, namely Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Qwen2.5-7B-Instruct, and Llama-3.2-3B-Instruct. This design choice was made to enable a comparative analysis across different LLM families, allowing us to evaluate variations in reasoning quality, alignment, and computational efficiency within a unified agentic workflow. Multi-agent systems depend on stable tool invocation, structured message passing, and deterministic orchestration across agents, making it essential to maintain consistent prompts and execution pipelines across all models.

While incorporating models from different families introduces heterogeneity, we ensured a standardized evaluation setup to allow fair comparison without extensive framework-specific modifications. Models such as GPT-4o [44] were not included due to access constraints and usage limitations, which would hinder systematic experimentation under consistent evaluation conditions. Therefore, the comparison in this work focuses on analyzing reasoning performance and latency trade-offs across multiple LLM families within a controlled and uniform framework.

To contextualize the proposed architecture, a single-agent RAG baseline would consist of a unified agent that retrieves relevant transcript chunks from the FAISS vector store and generates responses directly from the retrieved context. While such a setup is effective for factual question answering, it lacks the ability to perform structured, multi-step reasoning required for tasks such as tone analysis, temporal sentiment shifts, investment reasoning, and simulation. In contrast, the FARE framework decomposes these tasks into specialized agents with dedicated responsibilities, enabling more robust reasoning, improved interpretability, and better alignment with complex financial queries, while introducing additional architectural complexity compared to a unified single-agent setup.

B. SPECIALIZED AGENTS

The specialized agents of FARE operate independently of each other for the tasks designed for them. Our framework aims to capture the various user requirements in a question-and-answer style through these agents. Earlier financial agent frameworks were designed to achieve either trading insights, investment advice or tone analysis; however, FARE aims to combine all of these along with a robust evaluation layer to enhance the overall framework. The following sections describe each of the six specialized agents.

1) TONE AGENT

The Tone Agent is designed to analyze the tone of the earnings call transcripts. Figure 4 illustrates the workflow of the Tone Agent. A user will ask the orchestrator, "What is the tone of Company X in Quarter Y, Z," where X, Y, and Z are variables representing the company name, quarter, and year. The Tone Agent will only focus on the chunks for the specific quarterly report asked by the user. Since there are

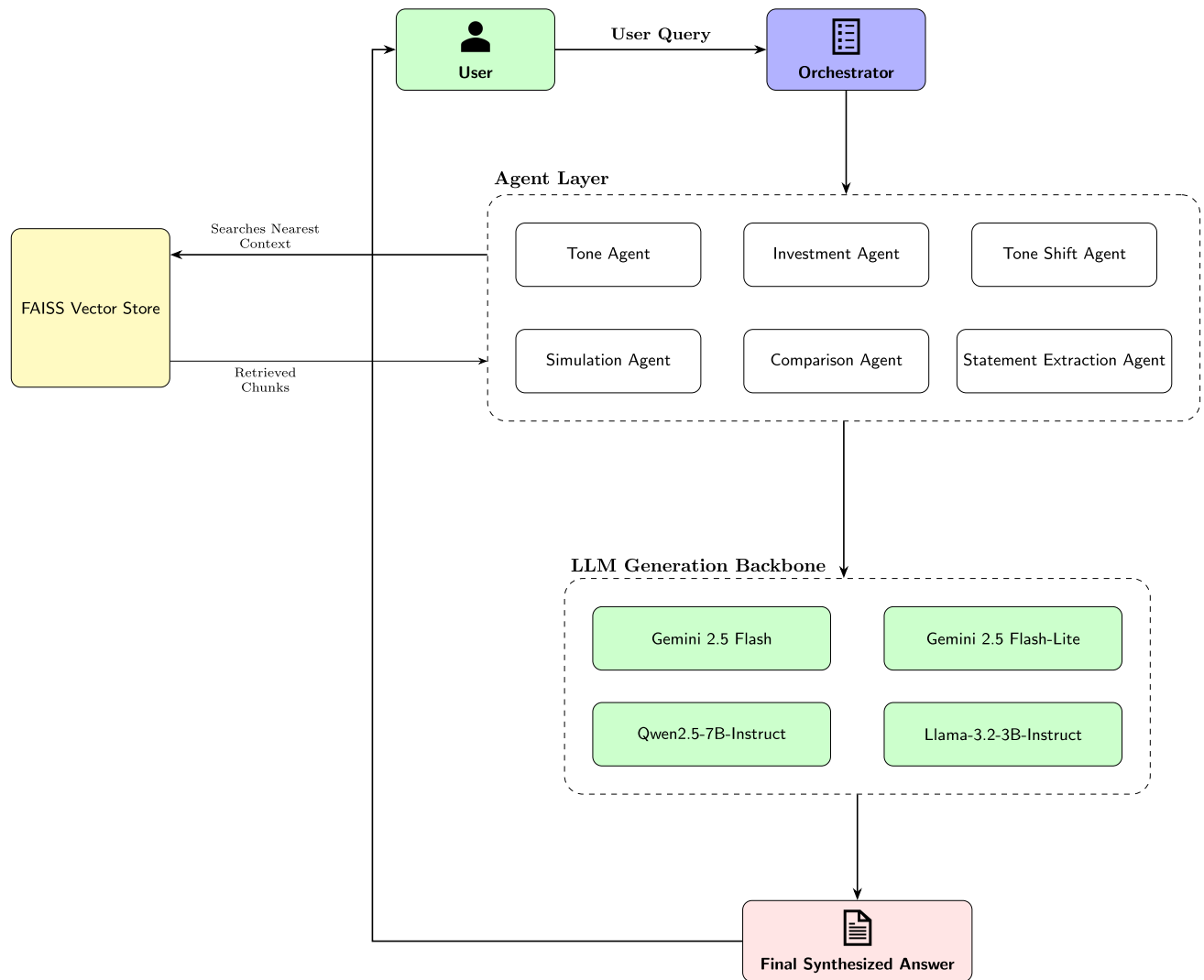


FIGURE 3. Workflow of the proposed FARE framework.

many chunks, the agent analyzes them in a batch-wise manner to ensure that the context window is not too large. Rather than specifying only an overall tone, the agent's goal is to identify nuances across different sections of the transcript. For example, a company may have its AI sector booming, which would indicate a positive sentiment, while its overall cash flow may have dropped, which would indicate a negative sentiment. The tones can vary from confident, optimistic, cautious, mixed, or uncertain.

After the Tone Agent outputs the initial response, a Tone Critic Agent evaluates the response from the Tone Agent. The Critic Agent calls the Google Search tool, which is used for grounding the answers produced by the Gemini models. This tool retrieves the ground truth regarding the tone that analysts have assigned to that company during that quarter. The Tone Critic Agent then provides feedback to the Tone Agent, asking it to make changes to the initial tone assessment. The

Tone Critic Agent is used exclusively during the evaluation phase and is not involved in the inference-time response generation. Our framework utilizes in-context learning for the feedback loop of the Tone Agent by providing the Tone Agent with an example of the correct response and how it should correct the initial response based on the example. The one-shot learning helps the Tone Agent provide a revised and balanced response that ensures the tonal judgment aligns with market understanding.

2) INVESTMENT AGENT

The Investment Agent is designed to provide investment advice to the user. Figure 5 illustrates the workflow of the Investment Agent. First, the user asks whether he made the right decision investing in a certain company, such as Microsoft or Accenture. The user also provides a user portfolio as context. The portfolio contains all the companies

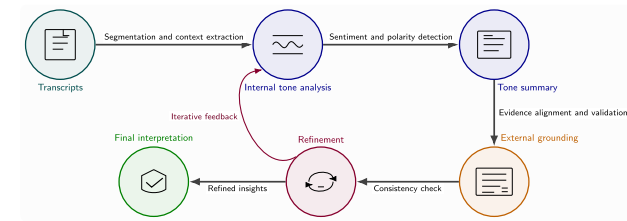


FIGURE 4. Workflow of the tone agent illustrating internal tone analysis, external evaluation through grounding, and refinement via an iterative feedback loop leading to the final interpretation.

in which the user has invested, the user's risk profile (light, moderate, or high risk), the amount invested, the reason for the investment, and the date on which the investment was made.

Based on all this information, the Investment Agent will identify the correct quarterly report relevant to the user's holdings. Then, it will analyze the financial performance of the company in a batch-wise manner and determine whether the original reason for the investment is still valid. Finally, the Investment Agent will provide a response summarizing the overall performance of the company and give a recommendation to the user. This recommendation will be one of the following: (1) hold the current investment, (2) re-evaluate the current investment, or (3) sell the investment.

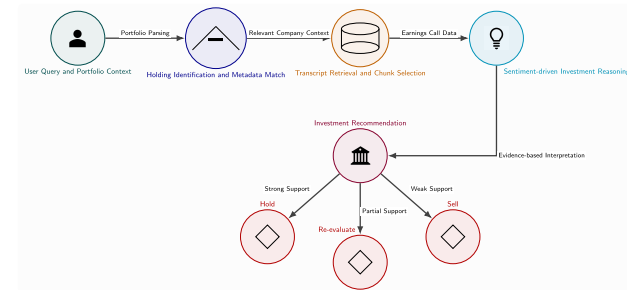


FIGURE 5. Workflow of the investment agent showing sentiment-driven reasoning leading from user query and transcript retrieval to evidence-based investment recommendations, which result in actionable decisions: hold, re-evaluate, or sell.

3) TONE SHIFT AGENT

The Tone Shift Agent is designed to analyze the change in sentiment of the overall performance of the company across multiple fiscal periods. Figure 6 illustrates the workflow of the Tone Shift Agent. Unlike the Tone Agent, which captures the sentiment of a single quarter, the Tone Shift Agent analyzes sentiment across multiple quarters.

First, the user asks the Orchestrator Agent to analyze the tone shift of a company and specifies the start and end quarters and years. The Orchestrator then routes this query to the Tone Shift Agent. The agent operates in a multi step manner, since there are multiple quarters and each quarter contains multiple chunks. First, it determines the relevant range and timeline that needs to be captured. Because the

user's query only mentions the start and end quarters, the agent has to infer the missing intermediate quarters. Then, it processes the data sequentially, starting from the initial quarter specified by the user, and analyzes each quarter in a batch wise manner. This process continues for all the quarters in the specified range. For each quarter, the agent infers the overall sentiment of the company, similar to the Tone Agent. Finally, the Tone Shift Agent combines the summaries of all quarters and outputs a tone evolution map of the company, which highlights the subtle reasons why the performance has changed and in which sectors. This analysis is crucial because the financial market is volatile, and FARE aims to bridge the gap between capturing not just static changes but also fluctuating trends over time.

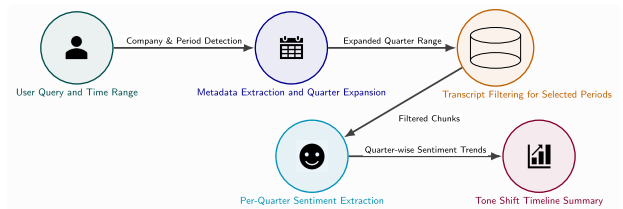


FIGURE 6. Workflow of the tone shift agent showing sequential reasoning from metadata extraction and quarterly expansion to multi-period sentiment analysis and tone evolution summarization.

4) SIMULATION AGENT

The Simulation Agent aims to model hypothetical portfolio reallocations in order to assess how the investment outcomes would have been different if the user had invested in another company. Figure 7 illustrates the workflow for the Simulation Agent. The Simulation Agent receives the user query and the user portfolio. As mentioned earlier, the user portfolio contains the companies in which the user has invested. The user query also contains another company, which becomes the hypothetical investment company.

The Simulation Agent will analyze in a batch wise manner the performance of both the actual and the hypothetical company. Since the portfolio contains the date on which the user invested, the agent must analyze the corresponding quarter for the hypothetical company as well. In our FARE framework, we also decided to make the agent analyze a few future quarters so that it can determine whether the hypothetical company is growing or stagnating. In this framework, 'future quarters' refers to the limited number of subsequent earnings-call periods following the initial investment quarter, typically ranging from two to four quarters based on the availability of the transcripts.

Because the user's investment depends on the current market outlook, and instead of providing the agent with today's market situation, the Simulation Agent must estimate future conditions by analyzing the trends observed across these future quarters. This trend-oriented forward reasoning performed by the agent allows it to identify directional signals such as improvements, stagnation, or deterioration

in the overall business outlook of the company. The agent does not perform time-series forecasting or quantitative price prediction. Rather, the agent will analyze the qualitative and quantitative information present in the earnings call transcripts to make an informed investment decision.

After the Simulation Agent is done analyzing, it outputs a three part response that includes: (1) whether the user missed an opportunity or avoided underperformance, (2) a narrative simulation of the alternative scenario, and (3) investment guidance that is aligned with the user's portfolio.

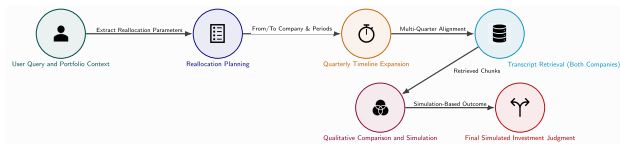


FIGURE 7. Workflow of the simulation agent showing sequential reasoning from reallocation planning and multi-quarter transcript retrieval to qualitative comparison and simulated investment judgment.

5) COMPARISON AGENT

The Comparison Agent is designed to compare the performance of different companies on a specific topic. Figure 8 illustrates the workflow of the Comparison Agent. The user query contains the specific quarter and the companies that must be compared. Similar to the previous agents, the Comparison Agent analyzes the quarterly transcripts for each company in a batch wise manner. Specifically, it reasons about whether the topic given by the user, which could be a specific market sector, is performing well or not. At the end of the analysis, the Comparison Agent provides a comparative summary and offers insights on the trends related to the topic for the different companies. This analysis is very important for the FARE framework, since users and stakeholders would want to understand how a sector of the market is performing across companies in order to gain an idea of which company to invest in.

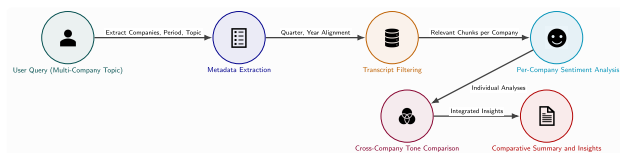


FIGURE 8. Workflow of the comparison agent showing metadata extraction, per-company sentiment evaluation, and synthesis into a comparative summary of managerial tone and strategic outlook.

6) STATEMENT EXTRACTION AGENT

The Statement Extraction Agent is designed to answer questions that are factual and involve specific details. Figure 9 illustrates the workflow of the Statement Extraction Agent. First, the user will ask a query such as “How is 3M trying to improve its innovation and product development process in Q3 2024?” The agent retrieves the relevant quarterly

transcript and locates the specific parts where the answer can be found. This is done through a semantic similarity search, where the most relevant chunks are identified. The agent then reasons through the chunks and provides a direct answer that includes the speaker who provided the information in the transcript and the actual quotes pertaining to the answer.

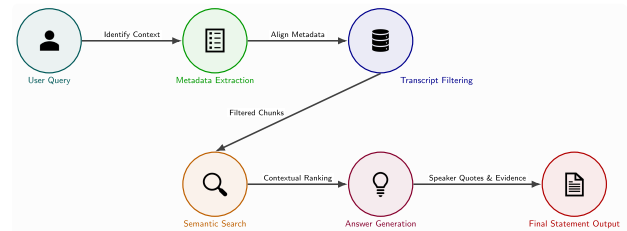


FIGURE 9. Workflow of the statement extraction agent showing metadata parsing, transcript filtering, semantic search, and evidence-grounded statement generation across two processing layers.

C. EVALUATION METRICS

Within the FARE framework, we evaluated each of the agents using qualitative metrics. Due to the lack of robust evaluation metrics in financial agentic AI systems, we introduced a set of novel metrics to evaluate our system. Each of these evaluation metrics was designed using an LLM-as-a-judge framework, where the LLM utilizes an external Google Search tool to retrieve financial analysis on the corresponding earnings call transcripts as a reference for evaluation. Although the LLM-as-a-judge framework enables scalable evaluation, it may introduce biases due to model priors, prompt sensitivity, and reliance on external retrieval grounding. However, the use of structured prompting and consistent evaluation settings helps maintain reasonable stability across assessments. The Google Search tool is used exclusively during the evaluation phase within the LLM-as-a-judge framework and is not utilized during inference or response generation by the agents. Unlike traditional evaluation metrics such as accuracy, F1-score, and BLEU, which assess syntactic and surface-structure correctness, the proposed metrics are designed to evaluate multi-step reasoning alignment and financial interpretability.

To ensure comprehensive validation, each metric captures a distinct aspect of agentic reasoning. Specifically, TAEF evaluates tonal alignment and interpretive consistency, IRAS assesses investment reasoning with respect to financial performance and risk alignment, TAS captures temporal consistency in sentiment evolution, EAS evaluates simulation-based reasoning and confidence calibration, and CAS measures the correctness of inter-company comparative analysis. Additionally, RAGAS-based metrics, including Context Precision, Faithfulness, and Response Relevancy, evaluate retrieval quality, factual grounding, and semantic alignment for the Statement Extraction Agent.

Collectively, these metrics provide a multi-dimensional evaluation framework that reflects the validity of the proposed solution across retrieval accuracy, reasoning

correctness, temporal consistency, financial alignment, and comparative reasoning.

1) TONE ALIGNMENT EVALUATION FRAMEWORK (TAEF)

The Tone Alignment Evaluation Framework (TAEF) provides a structured method to assess how effectively the Tone Agent aligns its refined tone analysis with both the company transcript and the external analyst commentary provided by the Tone Critic Agent. Figure 10 illustrates the workflow for the evaluation of the Tone Agent.

TAEF is based on three equally weighted evaluation dimensions, each scored on a scale from 0 to 2, where 0 indicates poor alignment, 1 indicates partial alignment, and 2 indicates strong alignment. A final normalized score in the range [0,1] is calculated from these three dimensions. The definition of these dimensions is shown below:

- **Integration of New Information (NI):** Evaluates how accurately and meaningfully the refined answer incorporates insights from analyst-grounded content.
- **Retention of Original Transcript Tone (OT):** Measures whether the refined output preserves the essential aspects of the original transcript such as the communicative intent, tone and polarity.
- **Nuance Awareness (NA):** Evaluated whether the model can recognize and express subtleties in tone, such as conditional optimism and hedging, when given the new perspective.

TAEF is calculated using Equation 1:

$$TAEF = \frac{NI + OT + NA}{6} \quad (1)$$

A higher TAEF score indicates that there was stronger tonal alignment, characterized by the effective integration of new information, faithful tone preservation, and nuanced interpretive understanding.

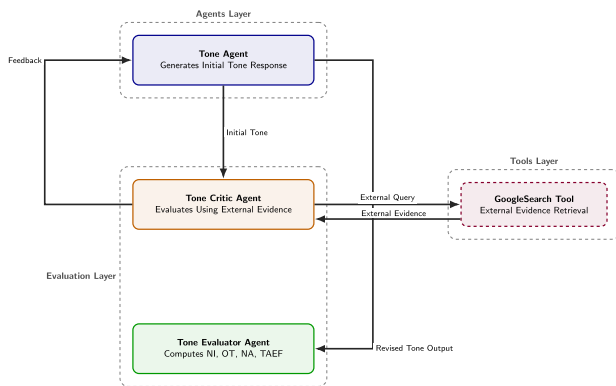


FIGURE 10. Evaluation workflow for the tone agent, including critique and refinement cycles using external evidence.

2) INVESTMENT REASONING ALIGNMENT SCORE (IRAS)

The Investment Reasoning Alignment Score (IRAS) evaluates how well the Investment Agent's recommendations are

aligned with real-world market outcomes, the investor's risk profile, and the logical validity of its reasoning. The IRAS score ranges from 0 to 1 and incorporates four evaluation dimensions. Each of these dimensions are scored on an ordinal scale from 0 to 1, where lower values indicate weaker alignment and higher values indicate stronger alignment. The evaluation dimensions are defined as follows:

- **Performance Correctness (PC):** Evaluates whether the investment outcome that is recommended aligns with real market performance. This is done using financial metrics such as Cumulative Return (CR), Annualized Return (AR), Sharpe Ratio (SR), and Maximum Drawdown (MDD).
- **Risk Profile Fit (RF):** Evaluates whether the investment outcome that is recommended is suitable for the investor's risk profile that he provided. Metrics such as Volatility and Maximum Drawdown are calculated relative to the investor's risk profile.
- **Confidence Calibration (CC):** Assesses whether the tone of the agent's recommendation (high, medium, or low confidence) appropriately reflects the actual investment outcome.
- **Reason Fidelity (RFid):** Evaluates whether the agent's response on whether the user's original reason for investment is valid. This is done by cross-checking against external market data.

The weighted IRAS is calculated using Equation 2:

$$IRAS = 0.35 \times PC + 0.25 \times RF + 0.20 \times CC + 0.20 \times RFid \quad (2)$$

Each component captures a distinct part of the evaluation, ensuring a comprehensive measure of reasoning quality. The score is interpreted using the following analysis:

- $IRAS > 0.7$: There is strong support — The recommendation aligns well with outcomes.
- $0.4 \leq IRAS \leq 0.7$: There is mixed support — The recommendation is partially aligned or is uncertain.
- $IRAS < 0.4$: Reject the agent's response — The recommendation is inconsistent with market data.

A higher IRAS value indicates that the agent's investment reasoning demonstrates a balanced factual accuracy, an appropriate tone, proper risk awareness, and reasoning fidelity.

3) TONE ALIGNMENT SCORE (TAS)

The Tone Alignment Score (TAS) is used to evaluate the Tone Shift Agent. It measures whether the agent's generated tone shift summary over multiple quarters is aligned with external analyst and news-based sentiment for the same time periods. The TAS score ranges from 0 to 1 and is computed based on four qualitative evaluation dimensions, each scored on an ordinal scale where higher values indicate stronger alignment. The dimensions are defined as follows:

- **Quarter-wise Match:** Evaluates whether the sentiment provided by the agent for each quarter matches with the real sentiment of the quarter from external evidence.
- **Tone Shift Trajectory:** Evaluates whether the agent's response when providing the tone shift analysis across the quarters is natural and coherent.
- **Evidence Usage:** Examines whether the agent uses relevant quotes and evidence in the response.
- **Confidence Appropriateness:** Evaluates whether the certainty that is expressed by the agent (high, medium, or low) is appropriate relative to the external sentiment.

The interpretation of TAS is shown as follows:

- $TAS > 0.9$: Full alignment, accurate tone shift response
- $0.7 \leq TAS \leq 0.9$: Strong alignment with minor mismatches
- $0.5 \leq TAS < 0.7$: Partial alignment with moderate inconsistencies
- $TAS < 0.5$: Weak or incorrect alignment

A higher TAS indicates that the Tone Shift Agent was able to effectively capture sentiment changes over time that is matched with external evidence.

4) EVALUATION ALIGNMENT SCORE (EAS)

The Evaluation Alignment Score (EAS) is in the range [0, 1] and is used to evaluate whether the Simulation Agent's investment recommendation aligns with real-world performance metrics and the investor's risk profile. It captures both factual consistency and confidence realism, offering a holistic way to validate the quality of simulated investment reasoning. The evaluation utilizes the following three dimensions:

- **Performance Correctness:** Evaluates whether the simulated recommendation aligns with real financial metrics such as Cumulative Return (CR), Annualized Return (AR), Sharpe Ratio (SR), and Maximum Drawdown (MDD).
- **Risk Alignment:** Evaluates whether the simulated outcome is matches with the investor's risk profile, based on observed volatility and drawdown.
- **Confidence–Outcome Divergence Score (CODS):** Evaluates how well the agent's expressed confidence matches with the correctness of its recommendation. Score is on a 1–5 scale (1: severe divergence, 5: perfect alignment).

The interpretation of EAS is shown as follows:

- $EAS > 0.7$: There is Strong Support. The performance metrics, risk alignment, and confidence aligns with the agent's claim.
- $0.4 \leq EAS \leq 0.7$: There is Mixed Support. Some metrics align but others indicate uncertainty.
- $EAS < 0.4$: Reject the agent's response due to inconsistency with real metrics and investor risk profile.

A higher EAS value signifies that the Simulation Agent's recommendation was able to demonstrate both factual accuracy and appropriate confidence calibration, indicating an overall strong reasoning alignment (RA).

5) COMPARISON ALIGNMENT SCORE (CAS)

The Comparison Alignment Score (CAS) is used to evaluate the Comparison Agent. It measures how accurately the agent compares multiple companies on a single topic within a specified quarter and year, and whether the comparison matches with external evidence. The CAS is normalized between 0 and 1 and is computed based on two evaluation dimensions:

- **Factual Alignment (FA):** Evaluates how well the agent's statements about each company match with external analyst evidence. Each company i receives a score between 0 and 2.
- **Reasoning Alignment (RA):** Evaluates the logical validity of the overall comparative conclusion drawn across all companies for the given topic. A score between 0 and 2 is assigned.

The final Comparison Alignment Score is calculated using Equation 3:

$$CAS = \frac{\sum_{i=1}^M FA_i + RA}{2M + 2} \quad (3)$$

Here, $2M$ represents the maximum achievable points for factual alignment (FA), which is two points per company. The additional 2 accounts for the reasoning alignment (RA) component. A higher CAS shows that the Comparison Agent's response matches with the external evidence.

6) STATEMENT EXTRACTION RAG METRICS

The Statement Extraction Agent retrieves factual statements grounded in context from the earnings call transcripts. Its performance is evaluated using three metrics from the RAGAS framework, each normalized on a [0,1] scale [13].

- **Context Precision (CP):** Evaluates how effectively the retriever ranks relevant chunks higher than irrelevant ones for a given query. It measures the degree to which relevant context appears near the top of the retrieved results. It is computed using Equation 4.

$$CP@K = \frac{\sum_{k=1}^K (P@k \times v_k)}{R_K} \quad (4)$$

where $P@k$ is the precision at rank k , $v_k \in \{0, 1\}$ is the relevance indicator for the chunk at rank k , and R_K is the number of relevant items among the top K retrieved chunks. $P@k$ is computed using Equation 5.

$$P@k = \frac{TP@k}{TP@k + FP@k} \quad (5)$$

where TP and FP denote true and false positives, respectively.

- **Faithfulness (F):** Measures how factually consistent the generated statement is with the retrieved context. A statement is considered faithful if all its claims can be supported by the context. It is computed using Equation 6.

$$F = \frac{C_s}{C_t} \quad (6)$$

where C_s is the number of claims in the response supported by the retrieved context, and C_t is the total number of claims in the response. Higher scores indicate stronger factual grounding and fewer hallucinations.

- **Response Relevancy (RR):** Assesses how semantically relevant the generated statement is to the user's query. It checks whether the answer directly addresses the question without redundancy. It is calculated using Equation 7.

$$RR = \frac{1}{N} \sum_{i=1}^N \frac{E_{g_i} \cdot E_o}{\|E_{g_i}\| \|E_o\|} \quad (7)$$

where E_{g_i} represents the embedding of the i^{th} generated question, E_o represents the embedding of the user query, and N is the number of generated questions (default: 3).

V. RESULTS

This section presents the experimental results comparing the four models, namely Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Qwen2.5-7B-Instruct, and Llama-3.2-3B-Instruct, across all agents in the proposed FARE framework. Each agent is evaluated using the previously defined metrics and latency to assess the reasoning performance and computational efficiency. Table 2 provides the distribution of evaluation queries.

A. STATEMENT EXTRACTION AGENT

The Statement Extraction Agent was evaluated using RAGAS metrics such as Context Precision, Faithfulness, and Response Relevancy. These metrics measure retrieval accuracy, factual consistency, and semantic alignment. We also recorded latency of each agent to assess real-time usability. The quantitative results are summarized in Table 3.

As shown in Table 3, all models demonstrate strong factual grounding and semantic alignment, although with varying trade-offs. The Gemini models achieve the highest Faithfulness scores, indicating that their generated statements are highly consistent with the retrieved context. In contrast, Qwen2.5-7B-Instruct and Llama-3.2-3B-Instruct exhibit lower Faithfulness scores of 0.82 and 0.83 respectively, suggesting a higher tendency toward minor inconsistencies or unsupported claims.

In terms of Response Relevancy, Qwen2.5-7B-Instruct achieves the highest score of 0.84, followed closely by Llama-3.2-3B-Instruct, indicating that these models produce responses that are more directly aligned with the user query. The Gemini models, while slightly lower, still maintain strong semantic relevance, demonstrating a balanced trade-off between factual grounding and query alignment.

All models show relatively similar Context Precision values ranging from 0.47 to 0.52, indicating that a portion of retrieved chunks are not directly utilized in the final answer. This behavior is expected in earnings call transcripts, where relevant information is distributed across multiple segments rather than concentrated in a few highly relevant passages.

This is further influenced by the use of FAISS for dense retrieval, which retrieves semantically similar chunks that may not always be strictly relevant. Additionally, the framework retrieves the top-10 chunks to ensure sufficient contextual coverage for downstream reasoning. While reducing this number could improve Context Precision, it may negatively impact answer completeness and Faithfulness. Therefore, the chosen configuration reflects a trade-off between retrieval precision and maintaining adequate context for accurate response generation.

From an efficiency perspective, Gemini 2.5 Flash-Lite achieves the lowest latency of 9.11 s, significantly outperforming all other models, while also maintaining high Faithfulness and competitive Response Relevancy. Gemini 2.5 Flash, although slower, provides similarly strong factual consistency. In contrast, Qwen2.5-7B-Instruct exhibits the highest latency of 52.25 s, despite achieving the best response relevancy, while Llama-3.2-3B-Instruct demonstrates the lowest overall performance.

Overall, Gemini 2.5 Flash-Lite provides the best balance between factual reliability, response quality, and computational efficiency, making it the most suitable model for real-time statement extraction within the FARE framework.

B. INVESTMENT AGENT

The Investment Agent was evaluated using the Investment Reasoning Alignment Score (IRAS) and Latency to assess its ability to produce logically coherent, risk-consistent, and performance-aligned investment recommendations. The IRAS metric quantitatively integrates four components, including Performance Correctness, Risk Profile Fit, Confidence Calibration, and Reason Fidelity, to measure the overall reasoning alignment between the model's recommendations, real market outcomes, and the investor's stated risk tolerance. The quantitative results are summarized in Table 4.

As shown in Table 4, all models achieved moderate IRAS scores, indicating mixed reasoning alignment across investment recommendations. Qwen2.5-7B-Instruct obtained the highest IRAS score of 0.63, followed closely by Gemini 2.5 Flash-Lite with 0.61, while Gemini 2.5 Flash achieved a slightly lower score of 0.53. In contrast, Llama-3.2-3B-Instruct recorded the lowest IRAS score 0.39, suggesting weaker alignment with market outcomes and investor risk profiles.

These results indicate that, on average, the models produced partially aligned recommendations, where the agent's output matched real market performance and risk considerations only to a moderate extent. This behavior is expected because the agent relies solely on transcript-based information and does not incorporate real-time financial data. Additionally, variations in risk profile alignment and confidence calibration contributed to inconsistencies in reasoning quality across models.

From an efficiency perspective, Gemini 2.5 Flash-Lite demonstrated significantly lower latency of 60.99 s compared to Gemini 2.5 Flash, while also achieving a higher IRAS

TABLE 2. Summary of query distribution used for evaluating each specialized agent.

Agent	Number of Queries	Query Sampling Description
Statement Extraction Agent	105	5 random quarters selected for each of the 21 companies (no sampling bias).
Investment Agent	21	7 randomly selected companies, each tested across all risk-profile variations.
Comparison Agent	200	Random combinations of company pairs and topics; all companies included across selections.
Tone Agent	105	5 random quarters chosen for each company (same sampling method as Statement Extraction).
Tone Shift Agent	63	3 queries per company: spanning 2 quarters, 3 quarters, and 4 quarters respectively.
Simulation Agent	50	10 randomly selected invested companies, each paired with 5 hypothetical alternatives.

TABLE 3. Performance evaluation for the statement extraction agent.

Model	Context Precision	Faithfulness	Response Relevancy	Latency (s)
Gemini 2.5 Flash	0.52	0.96	0.71	21.74
Gemini 2.5 Flash-Lite	0.52	0.99	0.73	9.11
Qwen2.5-7B-Instruct	0.48	0.82	0.84	52.25
Llama-3.2-3B-Instruct	0.47	0.83	0.82	27.66

TABLE 4. Performance evaluation for the investment agent.

Model	IRAS	Latency (s)
Gemini 2.5 Flash	0.53	182.55
Gemini 2.5 Flash-Lite	0.61	60.99
Qwen2.5-7B-Instruct	0.63	609.15
Llama-3.2-3B-Instruct	0.39	380.43

score. Qwen2.5-7B-Instruct, despite achieving the best IRAS performance, exhibited substantially higher latency of 609.15 s, limiting its practical usability. Similarly, Llama-3.2-3B-Instruct showed both lower reasoning performance and high latency, making it less suitable for real-time applications.

Overall, Gemini 2.5 Flash-Lite provides the best balance between reasoning alignment and computational efficiency, making it the most suitable model for delivering practical and responsive investment recommendations within the FARE framework.

C. COMPARISON AGENT

The Comparison Agent was evaluated using the Comparison Alignment Score (CAS) and Latency to measure its ability to perform inter-company comparisons based on factual consistency and logical reasoning coherence. CAS quantifies how accurately the agent contrasts multiple companies on a shared topic within a given quarter and year, balancing factual alignment with reasoning validity. The quantitative results are summarized in Table 5.

As shown in Table 5, all models demonstrate varying capabilities in performing inter-company comparisons. Llama-3.2-3B-Instruct achieves the highest CAS score of 0.77, followed by Qwen2.5-7B-Instruct with 0.71, indicating stronger reasoning alignment and factual consistency in comparative analysis. The Gemini models achieve slightly lower but comparable CAS scores, with Gemini 2.5 Flash

TABLE 5. Performance evaluation for the comparison agent.

Model	CAS	Latency (s)
Gemini 2.5 Flash	0.65	166.67
Gemini 2.5 Flash-Lite	0.64	14.30
Qwen2.5-7B-Instruct	0.71	698.67
Llama-3.2-3B-Instruct	0.77	485.75

at 0.65 and Gemini 2.5 Flash-Lite at 0.64, suggesting that they are able to perform effective comparisons, although with slightly less detailed reasoning than the other models.

These results indicate that while all models are capable of identifying broad comparative trends across companies, differences arise in the depth of reasoning and the ability to capture nuanced factors influencing performance. In particular, the higher CAS scores of Qwen2.5-7B-Instruct and Llama-3.2-3B-Instruct suggest stronger logical consistency, whereas the Gemini models provide more balanced but slightly less detailed comparative insights. This limitation is expected, as earnings call transcripts often contain implicit signals and contextual nuances that require deeper reasoning to fully interpret.

From an efficiency perspective, Gemini 2.5 Flash-Lite significantly outperforms all other models in terms of latency with 14.30 s, compared to Gemini 2.5 Flash with 166.67 s, Qwen2.5-7B-Instruct with 698.67 seconds, and Llama-3.2-3B-Instruct with 485.75 seconds. Although Qwen and Llama achieve higher CAS scores, their substantially higher latency limits their practicality for real-time applications.

Overall, Gemini 2.5 Flash-Lite offers the best trade-off between reasoning performance and computational efficiency, making it the most suitable model for real-time inter-company comparison tasks within the FARE framework.

D. TONE AGENT

The Tone Agent was evaluated using four primary metrics, namely NI, OT, NA, and TAEF, along with latency. These metrics collectively assess the agent's ability to capture tonal nuances and refine the initial tone response correctly. The quantitative results are summarized in Table 6.

As shown in Table 6, all models demonstrate strong performance across the NI, OT, NA, and TAEF metrics,

TABLE 6. Performance evaluation for the tone agent.

Model	NI	OT	NA	TAEF	Latency (s)
Gemini 2.5 Flash	1.93	2.00	1.99	0.99	148.02
Gemini 2.5 Flash-Lite	1.98	1.99	1.99	0.99	18.15
Qwen2.5-7B-Instruct	2.00	2.00	2.00	1.00	495.49
Llama-3.2-3B-Instruct	1.89	1.67	1.83	0.90	365.70

indicating effective tone understanding and refinement capabilities. Qwen2.5-7B-Instruct achieves perfect scores across all components, reflecting its ability to fully integrate external information, retain the original transcript tone, and capture nuanced sentiment. The Gemini models also achieve near-perfect performance, with both Gemini 2.5 Flash and Gemini 2.5 Flash-Lite attaining a TAEF score of 0.99, indicating highly accurate tone refinement and strong alignment with both internal and external context.

Llama-3.2-3B-Instruct, while still demonstrating strong performance, achieves comparatively lower scores, suggesting minor limitations in capturing nuanced tone shifts and preserving contextual alignment. Overall, these results indicate that all models are capable of effectively analyzing and refining tone, with differences primarily arising in the degree of nuance awareness and consistency.

From an efficiency perspective, Gemini 2.5 Flash-Lite significantly outperforms all other models with the lowest latency of 18.15 s. Although Qwen2.5-7B-Instruct achieves the highest evaluation scores, its substantially higher latency limits its practicality for real-time applications.

Overall, Gemini 2.5 Flash-Lite provides the best balance between near-perfect tone evaluation performance and computational efficiency, making it the most suitable model for the Tone Agent within the FARE framework.

E. TONE SHIFT AGENT

The Tone Shift Agent was evaluated using the Tone Alignment Score (TAS) and Latency to measure its ability to identify and interpret tonal continuity across consecutive quarters. TAS quantifies how accurately the agent's generated tone-shift summaries align with external analyst sentiment and evidence-based tone patterns over time. The quantitative results are summarized in Table 7.

TABLE 7. Performance evaluation for the tone shift agent.

Model	TAS	Latency (s)
Gemini 2.5 Flash	0.55	110.63
Gemini 2.5 Flash-Lite	0.67	30.92
Qwen2.5-7B-Instruct	0.87	191.21
Llama-3.2-3B-Instruct	0.75	126.53

As shown in Table 7, all models demonstrate varying effectiveness in capturing tonal continuity across quarters. Qwen2.5-7B-Instruct achieves the highest TAS score of 0.87, followed by Llama-3.2-3B-Instruct with 0.75, indicating stronger alignment with external analyst sentiment and a better ability to model tone progression over time.

These results indicate that while all models can identify general tonal trends, differences arise in their ability to model consistent tone trajectories across multiple periods. The higher TAS scores of Qwen2.5-7B-Instruct and Llama-3.2-3B-Instruct suggest more accurate temporal reasoning, whereas the Gemini models provide comparatively moderate alignment with external sentiment patterns. This limitation is expected, as modeling tone evolution requires integrating information across multiple transcript segments and maintaining temporal consistency.

From an efficiency perspective, Gemini 2.5 Flash-Lite significantly outperforms all other models with the lowest latency of 30.92 s. Although Qwen2.5-7B-Instruct achieves the highest TAS score, its higher latency reduces its practicality for real-time applications.

Overall, Gemini 2.5 Flash-Lite provides the best balance between temporal reasoning performance and computational efficiency, making it the most suitable model for tone evolution analysis within the FARE framework.

F. SIMULATION AGENT

The Simulation Agent was evaluated using the Confidence-Outcome Divergence Score (CODS), the Evidence Alignment Score (EAS), and Latency. These metrics collectively measure how confident and accurate the agent's response is and how well these simulations align with the real-world market evidence and risk profile of the user. The quantitative results are summarized in Table 8.

TABLE 8. Performance evaluation for the simulation agent.

Model	CODS	EAS	Latency (s)
Gemini 2.5 Flash	3.74	0.68	107.17
Gemini 2.5 Flash-Lite	3.28	0.64	53.68
Qwen2.5-7B-Instruct	4.08	0.76	189.37
Llama-3.2-3B-Instruct	2.60	0.70	101.65

As shown in Table 8, all models demonstrate varying performance in simulation-based investment reasoning. Qwen2.5-7B-Instruct achieves the highest CODS of 4.08 and EAS of 0.76, indicating stronger confidence calibration and better alignment with real-world market evidence and risk considerations. Llama-3.2-3B-Instruct attains a moderate EAS score of 0.70 but a lower CODS of 2.60, suggesting relatively weaker confidence calibration despite reasonable alignment with performance metrics.

Among the Gemini models, Gemini 2.5 Flash achieves a higher CODS and EAS compared to Gemini 2.5 Flash-Lite.

This indicates that Gemini 2.5 Flash is more consistent in aligning its simulated recommendations with actual market evidence and risk factors, while Gemini 2.5 Flash-Lite demonstrates slightly lower performance in these aspects. These results suggest that while both Gemini models provide reasonable simulation quality, they exhibit limitations in fully capturing complex risk-performance relationships inherent in financial decision-making.

From an efficiency perspective, Gemini 2.5 Flash-Lite significantly outperforms all other models with the lowest latency of 53.68 s. Although Qwen2.5-7B-Instruct achieves the highest evaluation scores, its higher latency reduces its practicality for real-time applications.

Overall, Gemini 2.5 Flash-Lite provides the best trade-off between acceptable simulation performance and computational efficiency, making it the most suitable model for real-time scenario analysis within the FARE framework.

G. QUALITATIVE ANALYSIS OF SAMPLE AGENT OUTPUTS

This section presents a qualitative analysis of representative outputs generated by each of the six specialized agents in the FARE framework. These examples illustrate how the system handles different types of financial queries, including factual extraction, tone analysis, temporal sentiment shifts, investment reasoning, simulation, and comparative analysis.

All sample outputs shown in this section are generated using Gemini 2.5 Flash-Lite as the LLM backbone to ensure consistency and demonstrate the framework's performance under a low-latency and efficient configuration. The outputs highlight the reasoning capabilities of each agent, along with their ability to utilize retrieved context and produce structured, interpretable responses. These outputs are intended to support financial reasoning and decision-making by providing structured insights from earnings call transcripts, rather than functioning as a fully autonomous investment recommendation system.

As shown in Figure 11, the Statement Extraction Agent effectively retrieves relevant transcript segments and generates a concise, evidence-grounded response to the user query. The retrieved top-*k* chunks capture key aspects of 3M's innovation strategy, including R&D allocation, product launches, and collaboration efforts. The generated answer synthesizes these insights into a coherent summary, while the supporting quotes provide transparency and traceability to the original transcript. The evaluation metrics further indicate strong factual grounding and high semantic alignment between the query and the generated response.

As shown in Figure 12, the Investment Agent evaluates the user's decision by combining transcript-based reasoning with performance-oriented financial validation. The generated response highlights key factors such as cash flow growth, content innovation, and revenue strategies to justify the investment decision. Additionally, the agent provides a structured recommendation along with relevant financial metrics, including cumulative return and risk indicators,

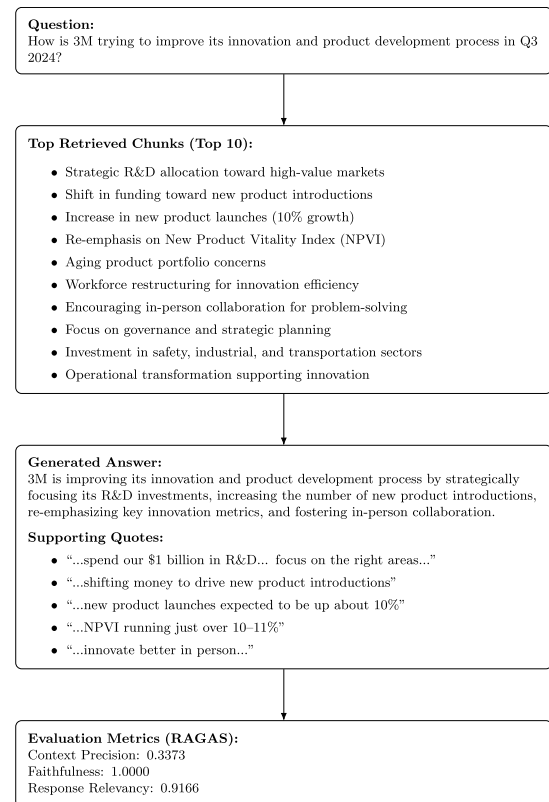


FIGURE 11. Sample output of statement extraction agent.

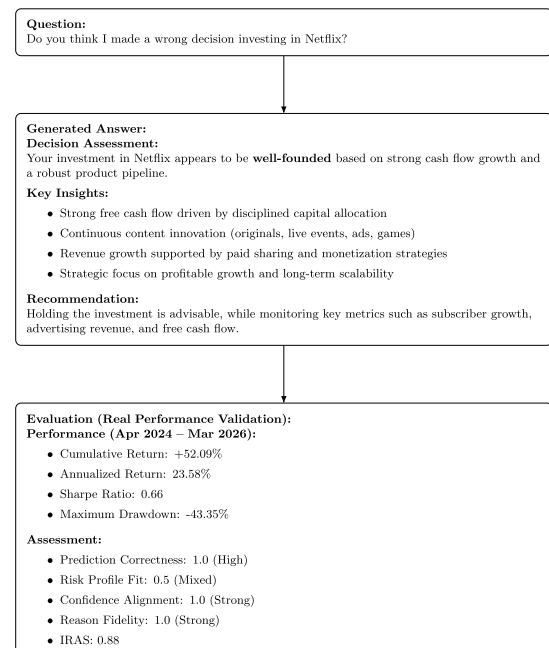


FIGURE 12. Sample output of investment agent.

to support its reasoning. This demonstrates the agent's ability to align qualitative insights from earnings call transcripts with quantitative performance measures. The high IRAS score further indicates strong reasoning alignment and consistency with real-world outcomes.

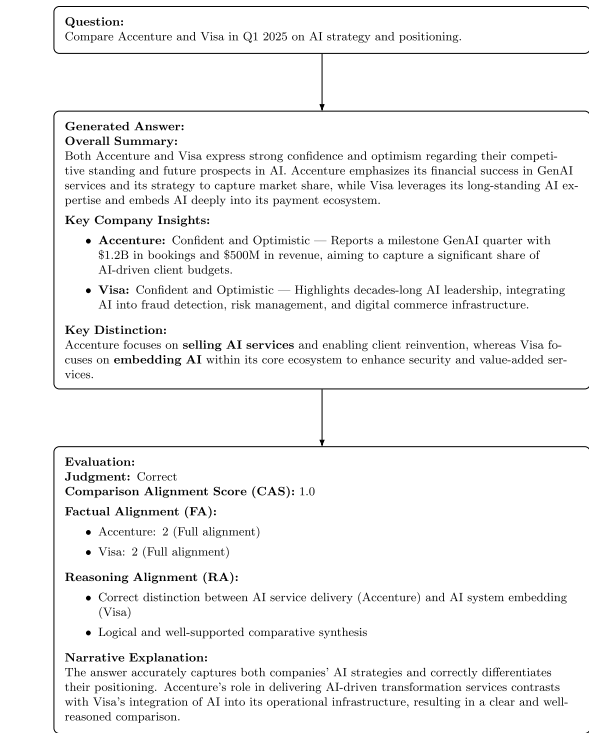


FIGURE 13. Sample output of comparison agent.

As shown in Figure 13, the Comparison Agent effectively analyzes and contrasts the AI strategies of Accenture and Visa using evidence from their respective earnings call transcripts. The generated response highlights both similarities, such as strong confidence in AI adoption, and key distinctions in strategic positioning between the two companies. Specifically, Accenture focuses on delivering AI-driven services, while Visa emphasizes embedding AI within its core payment infrastructure. The structured output, along with supporting reasoning, demonstrates the agent’s ability to perform coherent cross-company analysis. The high CAS score further indicates strong factual and reasoning alignment in the comparative assessment.

As shown in Figure 14, the Tone Agent effectively captures the nuanced sentiment of 3M’s earnings call by combining internal transcript analysis with external financial validation. The initial response identifies a balanced tone of optimism tempered with caution, which is further refined through the critic loop using analyst insights and market signals. The final output demonstrates improved alignment by incorporating external evidence such as financial performance and analyst upgrades. This iterative refinement highlights the agent’s ability to capture subtle sentiment variations rather than assigning a simplistic label. The perfect TAEF score further indicates strong alignment between the generated tone and external financial interpretation.

As shown in Figure 15, the Tone Shift Agent effectively captures the evolution of sentiment across consecutive quarters by analyzing changes in underlying risk factors. The generated response identifies a consistent tone of

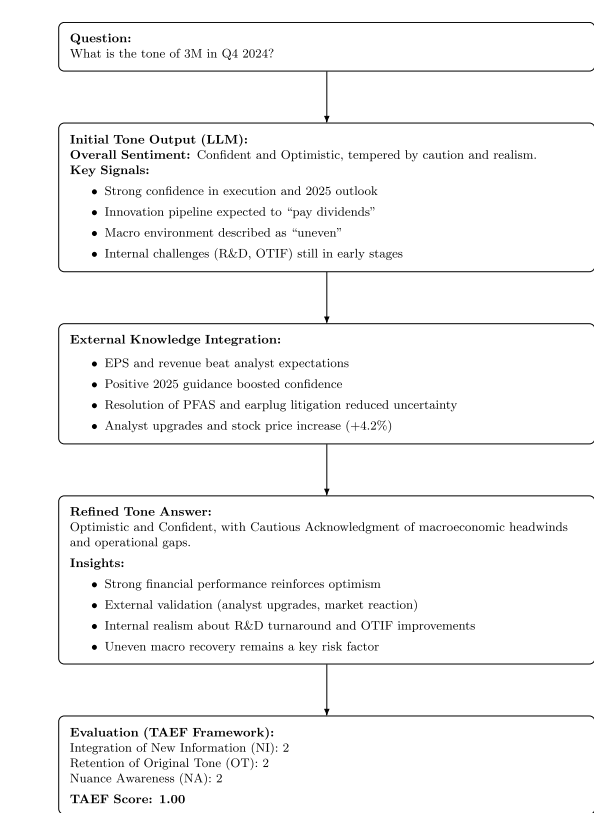


FIGURE 14. Sample output of tone agent.

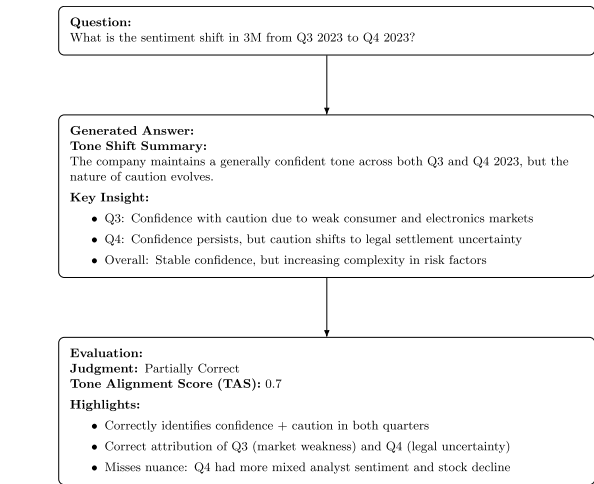


FIGURE 15. Sample output of tone shift agent.

confidence while highlighting shifts in the sources of caution, transitioning from market-related concerns to legal uncertainties. This demonstrates the agent’s ability to perform temporal reasoning and track changes in sentiment drivers rather than relying on static analysis. However, the evaluation indicates partial correctness, as some nuances in external analyst sentiment were not fully captured. The TAS score reflects this moderate alignment, indicating strong overall trend detection with minor inconsistencies in detailed interpretation.

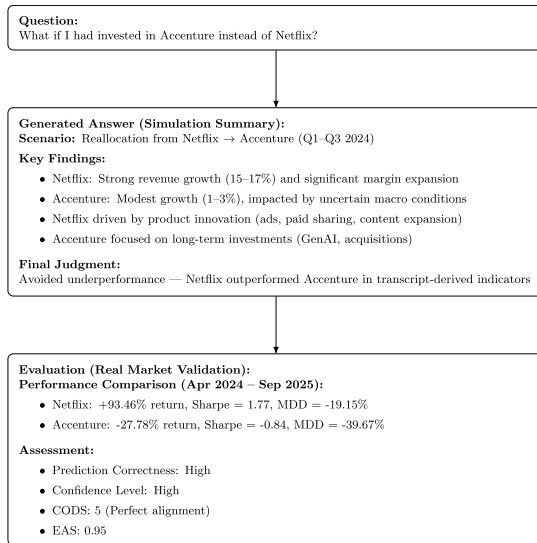


FIGURE 16. Sample output of simulation agent.

As shown in Figure 16, the Simulation Agent evaluates a hypothetical portfolio reallocation by comparing the performance and strategic direction of two companies. The generated response analyzes both qualitative transcript insights and quantitative performance indicators to assess the outcome of the alternative investment scenario. In this case, the agent correctly identifies that investing in Netflix resulted in superior performance compared to Accenture, supported by strong revenue growth and product innovation signals. The inclusion of real market validation metrics further strengthens the reliability of the simulation-based reasoning. The high EAS score and perfect CODS indicate strong alignment between the agent’s prediction and actual financial outcomes.

H. SUMMARY OF RESULTS

Figure 17 shows the latency comparison across all agents, where Gemini 2.5 Flash-Lite consistently achieves the lowest latency among all models. In contrast, Qwen2.5-7B-Instruct and Llama-3.2-3B-Instruct exhibit significantly higher latency across tasks. Figure 18 presents the efficiency comparison of the four models across all evaluation metrics used for the six specialized agents. Efficiency is defined in Equation 8.

$$\text{Efficiency} = \frac{\text{Evaluation Metric Score}}{\text{Latency}} \quad (8)$$

Gemini 2.5 Flash-Lite achieves the highest efficiency across all evaluation metrics, while Qwen2.5-7B-Instruct and Llama-3.2-3B-Instruct, despite strong performance in some tasks, are less efficient due to their higher latency. Therefore, Gemini 2.5 Flash-Lite offers the best trade-off between accuracy and speed, making it the most suitable model for practical deployment in the FARE framework.

I. COMPARISON WITH PREVIOUS STUDIES

Table 9 compares the proposed FARE framework with several existing multi-agent and retrieval-based approaches for financial document analysis. Prior systems such as ECC Analyzer, ARTI, FinDebate, and FinKPI each address specific sub-tasks, including volatility forecasting, ontology construction, debate-driven report synthesis, or KPI extraction. While these approaches demonstrate strong performance within their respective scopes, they do not provide a multi-agent reasoning framework capable of handling the wide range of financial reasoning tasks required for comprehensive earnings call analysis.

FARE differs from previous works in three ways. First, it introduces a multi-agent framework designed specifically for financial reasoning on earnings call transcripts, by using six specialized agents: Tone, Investment, Statement Extraction, Tone Shift, Comparison, and Simulation. Second, unlike prior work that focuses mainly on extraction or summarization, FARE incorporates a set of novel evaluation metrics that are tailored to financial agent behavior, such as TAEF, TAS, IRAS, EAS, and CAS, allowing fine-grained measurement of contextual alignment, reasoning integrity, and robustness. Third, FARE is designed for real-time usability by benchmarking multiple LLMs and prioritizing computational efficiency, which is rarely considered in previous studies.

Overall, FARE provides a broader perspective of financial document understanding by combining retrieval, domain-specific analysis, and multi-agent reasoning. This design enables practical deployment in real-time financial workflows, representing a significant advancement over existing systems that focus on isolated analytical capabilities.

VI. RISK ANALYSIS AND MITIGATION

The risk analysis in this study is conducted qualitatively, focusing on identifying key sources of error in financial agentic systems rather than assigning explicit probability or severity scores. The identified risks, including hallucination, interpretive inconsistency, and over-reliance on historical narratives, are considered high-impact due to their potential influence on financial decision-making. Accordingly, mitigation strategies are designed to directly address these risks through retrieval grounding, structured reasoning, and external evaluation. Despite these measures, residual risk remains in scenarios involving incomplete context or nuanced financial interpretation, and will be explored further in future work through formal risk quantification frameworks.

Financial agentic AI systems introduce new risks that include hallucinations, interpretive reliability issues, and the over-reliance on historical narratives. It is important for all users to be aware of such risks when using financial systems, and for all financial systems to incorporate appropriate risk mitigation strategies to ensure trust and transparency. The proposed framework FARE avoids hallucinations by incorporating a RAG pipeline for all agents to retrieve relevant

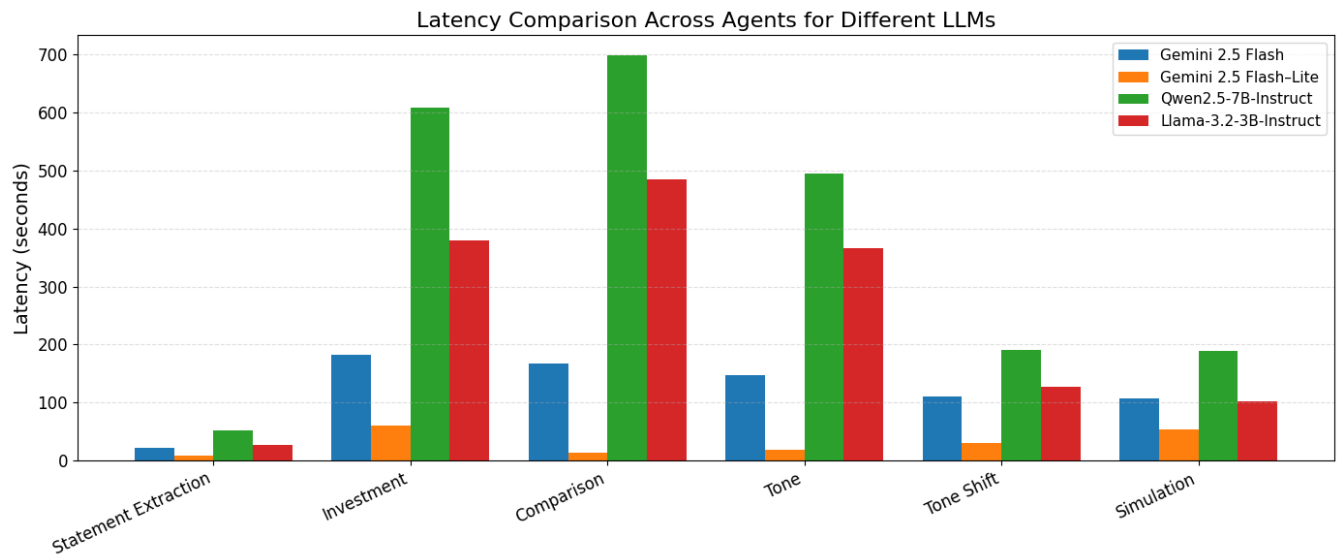


FIGURE 17. Latency comparison across all agents.

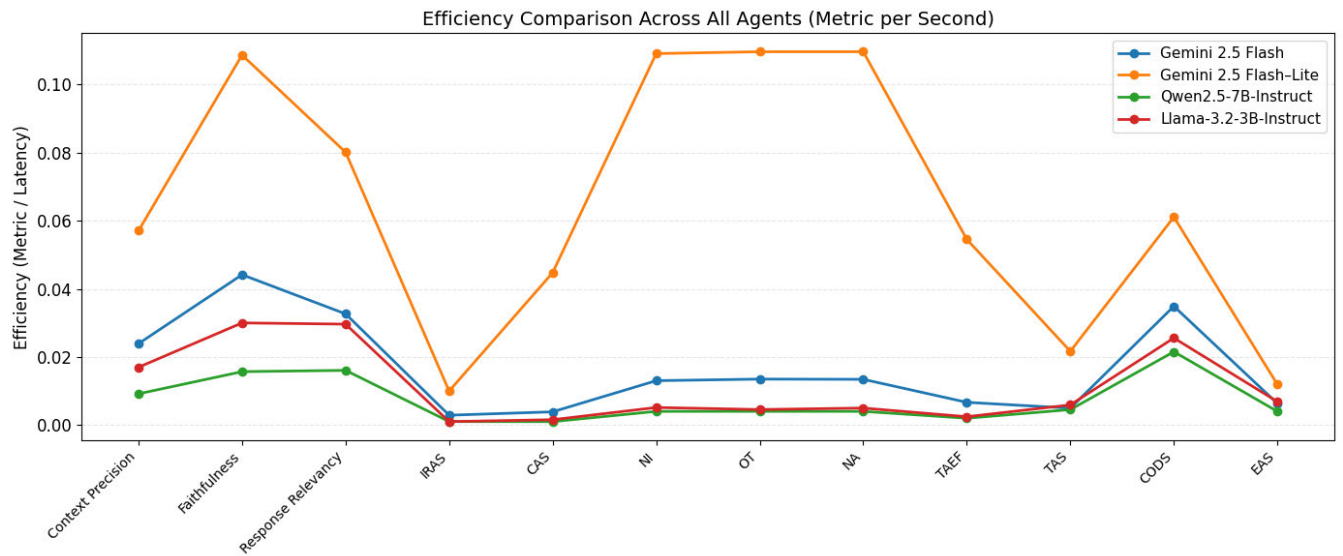


FIGURE 18. Efficiency comparison across evaluation metrics for all agents.

earnings call transcripts before answering user questions. Each agent is restricted to reason only over the retrieved transcript content, thereby reducing the likelihood of hallucinated responses. Prompts are specifically designed for all agents to ensure that reasoning is grounded exclusively in transcript evidence. Furthermore, robust evaluation is conducted on agent outputs using hallucination-related metrics such as Faithfulness.

Since FARE utilizes large language models, there exists a risk of misinterpretation of earnings call transcripts due to the substantial volume of textual information. Given the limited context window of LLMs, processing large amounts of text simultaneously may lead to incomplete or inaccurate interpretation of information. To address this issue, FARE ensures

that all agents analyze transcripts in a batch-wise manner, which enables a more focused and accurate interpretation of textual content. In addition, LLM-as-a-judge evaluation metrics are introduced to assess the interpretive capability of the agents by comparing their responses with reference financial analyses retrieved using the Google Search tool.

There is also a risk of over-reliance on historical narratives. To mitigate this, temporal scoping limits are enforced to ensure that agents reason only within the relevant earnings-call periods. Other risks, like potential conflicting signals, do not occur since the agents do not utilize real-time information to generate responses. Instead, real-time information is used only during the evaluation phase to assess whether the

TABLE 9. Comparison of FARE with existing multi-agent and retrieval-based financial analysis frameworks.

Approach	Key Components/Agents	Core Methodology	Dataset	Evaluation Metrics	Summary of Distinct Advantages and Limitations
ECC Analyzer [12]	Multimodal encoders (Wav2Vec2, SimCSE); summarization module; RAG with question bank; fusion layer.	Two-stage RAG: chunk summarization and query retrieval; fusion of audio, text, summary, and focus embeddings for volatility regression.	572 S&P 500 ECCs (2017); audio and text.	MSE for volatility prediction (3–30 days).	Predictive modeling focus; no agent coordination; narrower forecasting scope.
ARTI [45]	Topic Retriever, Topics Ontology, Ontologist Agent.	LLM-based topic extraction and ontology construction for trend evolution.	141 transcripts from 6 companies (EV/Semiconductors).	Ontology coherence; topic quality; trend significance.	Ontology evolution design; lacks investment/simulation reasoning; smaller dataset.
FinDebate [46]	5 domain agents (Analyst, Market, Sentiment, Valuation, Risk); report synthesizer; debate agents.	FinLang-based RAG; parallel report drafts with safe-debate refinement.	64 transcripts (ECTSum + analyst reports).	LLM coherence/accuracy scores; human evaluation.	Debate-based calibration; limited tone progression/comparison; no latency analysis.
FinKPI [47]	Extraction Agent; Text-to-SQL Agent.	Structured KPI extraction to JSON; NL-to-SQL querying with verification.	1,200 SEC filings; 312 analyst queries.	Extraction precision; SQL accuracy; unit error rate.	Structured KPI focus; lacks narrative reasoning, tone analysis, simulation.
FARE (Proposed)	6 agents (Statement Extraction, Tone, Investment, Comparison, Tone Shift, Simulation) plus an Orchestration Agent using LangGraph.	Multi-agent reasoning with RAG (FAISS + MiniLM embeddings); domain-specific analysis and synthesis; comparison of Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Qwen2.5-7B-Instruct, and Llama-3.2-3B-Instruct.	168 quarterly transcripts from 21 companies.	TAEF, TAS, IRAS, EAS, CAS; comprehensive alignment and robustness evaluation.	A financial multi-agent system integrating tone progression, simulation, comparison, and investment reasoning with real-time efficiency.

agent’s understanding and interpretation of the transcripts are sufficiently accurate.

By incorporating robust mitigation strategies for these risks, FARE ensures improved reliability, transparency, and interpretive trustworthiness. Users are able to observe the intermediate responses produced by each agent in real time, enabling transparency into how conclusions are formed and allowing users to retain full decision-making authority.

VII. LIMITATIONS AND FUTURE SCOPE

Despite the advantages of the proposed FARE framework, a few limitations exist. The framework does not include validation from financial experts for the proposed evaluation metrics. Additionally, the study used limited tools and did not utilize MCP servers. Furthermore, the risk analysis in this work is qualitative and does not include formal quantification of risk in terms of probability and severity, which may limit the granularity of risk assessment. Future enhancements will include:

- Direct validation of the proposed LLM-as-a-judge evaluation metrics with human financial experts, along with the incorporation of human prior knowledge into the evaluation and reasoning process, to strengthen the reliability and robustness of the framework, particularly in high-stakes investment scenarios.
- Use of the Agent Development Kit (ADK) along with integration of additional tools and MCP servers to develop a more scalable, expandable, and efficient system.
- Systematic ablation studies on various embedding models and chunking configurations, including dialogue-aware strategies based on speaker roles and turn-taking, to analyze their impact on retrieval relevance and context precision within the RAG pipeline of the framework.
- Development of formal risk quantification frameworks that incorporate probability–severity analysis and

structured risk scoring to enable a more systematic and rigorous assessment of system safety and reliability.

These improvements will make the proposed framework a more valuable asset for financial agentic systems.

VIII. CONCLUSION

This study proposed the Financial Agentic Reasoning and Evaluation (FARE) framework to enhance the interpretability and reliability of question answering over financial documents such as earnings call transcripts. The framework integrates six specialized agents, namely the Statement Extraction Agent, Tone Agent, Investment Agent, Comparison Agent, Tone Shift Agent, and Simulation Agent.

Compared to previous studies, we developed novel evaluation metrics to assess the performance of the agents. The metrics are the Tone Alignment Evaluation Framework (TAEF), Tone Alignment Score (TAS), Investment Reasoning Alignment Score (IRAS), Evaluation Alignment Score (EAS), and Comparison Alignment Score (CAS).

We compared the performance of four LLM models, namely Gemini 2.5 Flash, Gemini 2.5 Flash-Lite, Qwen2.5-7B-Instruct, and Llama-3.2-3B-Instruct, on the proposed framework. Qwen2.5-7B-Instruct outperformed the other models across majority of the agents, while Gemini 2.5 Flash Lite demonstrated superior efficiency across all the agents. FARE demonstrated a broad range of tasks that a multi agent system can perform, which are useful for investors, and provides transparent and factual answers that are grounded in analyst commentary.

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